



What Happens When an Automated System Makes the Wrong Decision — And There Is No Record?

OMNIX is the governance infrastructure that ensures every automated decision is evaluated, authorized, and cryptographically sealed — before it executes.

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\$137B+

Total Addressable Market

11

Governance Checkpoints per Decision

24/7

Production Operation — Live Now

Eureka GCC Dubai 2026
Semifinalist

SSRN · 2 Papers 6321298 ·
6507559

Zenodo · 2 DOIs 19056919 ·
19375792

Pre-Seed Round \$500K at \$3M
pre-money

The Real Cost of One Unaudited Decision

Every day, automated systems execute thousands of decisions that carry direct financial, legal, and reputational exposure. The question is not whether something will go wrong — it is whether you will be able to prove what happened when it does.

€35M

Maximum fine under EU AI Act per incident

\$50M+

Average regulatory fine for AML violations

78%

Of AI incidents leave no recoverable audit trail

"When regulators ask what the system decided, why it decided it, and who authorized it — most organizations cannot answer any of the three."

Scenario	Without OMNIX	With OMNIX
Automated trade executes incorrectly	No signed record. Dispute resolution takes months. Potential liability.	Signed receipt proves what was evaluated and authorized. Dispute resolved in hours.
Regulator requests AI decision audit	Logs may be incomplete or missing. No cryptographic proof. Non-compliance risk.	Every decision carries a tamper-evident receipt. Full audit trail delivered instantly.
Insurance claim denied by automated system	Client disputes denial. No verifiable record of evaluation criteria applied.	Receipt shows all 11 checkpoints passed or which one blocked the decision.
Autonomous system takes unauthorized action	No pre-execution gate. System acted without independent governance approval.	Fail-closed pipeline blocked execution. Receipt proves governance was applied.

What OMNIX Does

OMNIX sits between your automated system and the moment a decision executes. It evaluates every proposed decision through 11 independent governance checkpoints and issues a cryptographically-sealed receipt. No receipt — no execution.



If any checkpoint fails → pipeline halts → decision is BLOCKED → signed receipt issued.

Decision Result	What It Means	Receipt Issued?
APPROVED	All 11 checkpoints authorized the decision. Execution proceeds.	Yes — signed proof of authorization
BLOCKED	One or more checkpoints identified a governance violation. Execution halted.	Yes — signed proof of rejection and reason
HOLD	Internal signal contradiction detected. Human review required before proceeding.	Yes — signed escalation record

The 11 Governance Checkpoints

Each decision submitted to OMNIX passes through 11 independent checkpoints in sequence. Every checkpoint operates autonomously — a failure at any stage halts the pipeline immediately and issues a signed rejection receipt. No checkpoint can be bypassed or overridden.

CP	Checkpoint	What It Evaluates	If It Fails
CP -1	Signal Integrity Validator (SIV)	Verifies data quality, completeness, and input integrity before any evaluation begins.	Rejected at entry. Pipeline never opens.
CP -2	Probability Assessment	Evaluates statistical confidence in the proposed decision outcome against authorized thresholds.	Blocked. Insufficient confidence.
CP -3	Risk Evaluation	Quantifies downside exposure and compares against the authorized risk envelope for the domain.	Blocked. Risk limit exceeded.
CP -4	Coherence Engine	Detects internal contradictions across all active signals. Decision Contradiction Index (DCI) ≥ 70 mandates escalation.	HOLD. Human review required.
CP -5	Trend Validator	Confirms alignment between the proposed decision and the prevailing operational or market regime.	Blocked. Regime contradiction.
CP -6	Stress Testing	Simulates adverse conditions (liquidity shocks, volatility spikes) to validate decision robustness.	Blocked. Fails under stress.
CP -7	Ethics & Domain Gate	Enforces domain-specific ethical constraints: Sharia compliance (riba, gharar), safety limits for robotics, bias controls for credit.	Blocked. Ethics violation recorded.
CP -8	Threshold & Context Validator	Validates all decision parameters against authorized operational boundaries and contextual constraints.	Blocked. Parameter out of range.
CP -9	AML Screening	Screens for anti-money laundering indicators, suspicious transaction patterns, and sanctioned entity exposure.	Blocked. Suspicious activity flagged.
CP -10	Fraud Detection	Multi-layer fraud signal analysis across behavioral, transactional, and systemic patterns.	Blocked. Fraud flag escalated.
CP -11	Jurisdiction Compliance	Verifies regulatory jurisdiction eligibility before any cross-border or regulated decision executes.	Blocked. Jurisdiction violation.

"Every checkpoint is independent. The system cannot approve a decision because 10 out of 11 passed. All 11 must pass. This is what fail-closed governance means."

The Business Case: ROI & Value Delivered

OMNIX is not a cost center — it is a risk-reduction investment with a quantifiable return. The value delivered falls into three categories: regulatory protection, operational efficiency, and reputational capital.

\$500K+ Minimum regulatory fine avoided per incident	100% Audit trail coverage for every decision	<1ms Governance evaluation time per decision
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Value Category	Without OMNIX	With OMNIX	Estimated Impact
Regulatory Fines	No audit trail. Full exposure.	Signed receipt per decision.	€35M EU AI Act exposure eliminated
Incident Response	Weeks of log reconstruction.	Instant receipt retrieval.	80% reduction in response time
Legal Disputes	He said / she said. No proof.	Cryptographic proof of what was decided.	Disputes resolved in hours, not months
Compliance Reporting	Manual, error-prone, expensive to produce.	Automated, signed, instantly accessible.	60–80% reduction in compliance cost
Client Trust	Black box — cannot explain decisions.	Public verification URL per decision.	Measurable increase in client retention

Simple ROI Calculation

OMNIX Enterprise cost: \$20,000 – \$35,000 / month

One avoided EU AI Act fine: €35,000,000

Break-even point: 1 avoided incident = 145 years of OMNIX cost

The question is not whether OMNIX is expensive.

The question is whether you can afford to operate without it.

Four Active Governance Domains

The same governance pipeline operates across four distinct industries simultaneously. OMNIX does not need to be rebuilt for each domain — it is configured. This is what makes it infrastructure, not software.

Domain	What OMNIX Governs	Key Risk Eliminated	Status
Digital Asset Trading	Every trade entry and exit decision. Real capital on Kraken exchange.	Unauthorized execution under adverse conditions	LIVE 24/7
Islamic Credit GCC / MENA	Murabaha financing decisions. Sharia compliance enforced at CP-6.	Riba violation and non-compliant structuring	OPERATIONAL
Insurance Claims	Automated claim assessment and fraud screening decisions.	Fraudulent approval and unaudited denial	OPERATIONAL
Robotics Pre-Execution	Autonomous action authorization before any physical command executes.	Unauthorized physical action — safety breach	OPERATIONAL

"The same 11-checkpoint pipeline that governs a crypto trade also governs a Murabaha credit decision. Domain changes. Governance does not."

The Governance Receipt: Your Proof of Accountability

Every decision processed by OMNIX generates a receipt. This receipt is the evidence that governance was applied. It is cryptographically signed, tamper-evident, and independently verifiable by anyone — regulators, auditors, counterparties, or clients.

```
{
  "receipt_id": "omnix-gov-20260401-a7f3c9",
  "timestamp": "2026-04-01T14:32:17.841Z",
  "decision": "APPROVED",
  "checkpoints_passed": 11,
  "checkpoints_total": 11,
  "veto_count": 0,
  "governance_summary": "All checkpoints passed",
  "cryptographic_seal": {
    "algorithm": "PQC signature (OMNIX implementation)",
    "merkle_root": "a3f7c2d8e1b490fa231c...",
    "verification_url": "https://omnixquantum.net/verify/..."
  }
}
```

Field	What It Proves
receipt_id	Unique link to the complete decision audit record
decision	The governance determination: APPROVED, BLOCKED, or HOLD
checkpoints_passed	How many of the 11 checkpoints independently authorized the decision
cryptographic_seal	Post-quantum signature — mathematically impossible to forge or alter after issuance
verification_url	Public link — any third party can independently confirm this receipt at any time

Regulatory Protection Built In

Governance receipts are the evidence layer that regulators are beginning to require. OMNIX was designed with these frameworks as first-class requirements — not added as an afterthought.

Framework	Requirement	OMNIX Coverage
EU AI Act	High-risk AI systems must maintain decision logs and enable human oversight.	Every decision: immutable signed receipt + HOLD escalation for human review
NIST AI RMF	AI systems must be measurable, explainable, and subject to governance controls.	11 checkpoints provide structured, measurable evaluation. Receipts enable explainability.
ISO/IEC 42001	AI management systems require documented risk controls and incident traceability.	Fail-closed pipeline + public verification URL = full incident traceability
AAOIFI (Islamic)	Sharia-compliant products must adhere to riba and gharar prohibitions.	CP-6 enforces Sharia principles as a pre-execution governance gate

Why Not Just Use What You Already Have?

Every organization already has some form of logging, monitoring, or compliance tooling. The question is not whether those tools exist — it is whether they provide pre-execution governance with cryptographic accountability. They do not.

Alternative	What It Does	What's Missing	OMNIX Difference
Audit Logs	Record what happened after execution has already occurred.	Evidence created after the fact. Cannot prevent a bad decision.	OMNIX governs BEFORE execution. The receipt proves authorization, not reconstruction.
SIEM / Monitoring	Alert teams to anomalies once they are detected.	Cannot halt execution. Cannot produce signed proof of evaluation. Observation only.	OMNIX halts. Fail-closed by design. No signed receipt = no execution. Ever.
Compliance Platforms	Report policy adherence and flag violations after the fact.	Policy check ≠ governance gate. No cryptographic seal. Cannot be independently verified.	OMNIX enforces. Every decision carries a tamper-evident, publicly verifiable signed receipt.
Custom Internal Validation	Domain-specific code checks written by internal engineering teams.	Non-standardized. No independent verification. Cannot audit externally. Breaks across domains.	OMNIX is domain-agnostic infrastructure. Same pipeline, same receipt standard, across every vertical.
Human Review Committees	Manual oversight for high-stakes decisions by designated reviewers.	Does not scale. Inconsistent. No cryptographic record of what was reviewed and why.	OMNIX escalates to human review only when the Coherence Engine detects contradiction (CP-4 HOLD).

"The difference between an audit trail and governance infrastructure is the same as the difference between a black box recorder and a pilot. One tells you what happened. The other determines whether it should happen at all."

How to Start

OMNIX is designed to integrate without disrupting your existing systems. The Shadow Mode tier allows full governance evaluation in parallel with your current operations — zero risk, immediate visibility.

Tier	How It Works	Best For	Investment
Shadow Mode	OMNIX runs in parallel. Evaluates every decision. No intervention. Full receipt generation.	Organizations that want visibility before committing to full governance.	Complimentary
Advisory	OMNIX results delivered as recommendations. Your team retains final decision authority.	Regulated environments where human oversight is currently mandatory.	\$8,000 / month
Enterprise	Full integration. OMNIX receipt required before any automated decision executes. Fail-closed.	Organizations with significant regulatory exposure or high-stakes automated operations.	\$20,000 – \$35,000 / month
Custom	Multi-domain deployment, private infrastructure, custom governance parameters, SLA guarantees.	Financial institutions, government, and infrastructure operators at scale.	Contact us

Operational Track Record

OMNIX has been in continuous production operation since July 2025, with the governance pipeline actively evaluating decisions across all four domains. The system has maintained 24/7 uptime with zero infrastructure failures.

Note: Digital asset trading operates with real capital on Kraken exchange. Islamic credit, insurance, and robotics domains operate in a controlled validation environment — governance logic is identical; domain-specific parameters are under calibration for production deployment.

Milestone	Date	Significance
Production deployment	July 2025	Live operation with real capital on Kraken exchange. Governance pipeline validated under live market conditions.
PQC implementation	November 2025	Post-quantum cryptographic signature deployed. All receipts carry NIST-standardized PQC signatures.
Multi-domain expansion	March 2026	Islamic credit, insurance, and robotics domains activated in validation environment. Same pipeline across all four sectors.
Track Record phase	January 2026	Recalibrated pipeline activated. Official governance performance measurement period begins.
Academic publications	March 2026	4 peer-indexed publications: SSRN abstract_id=6321298 (PQC architecture) and 6507559 (governance whitepaper); Zenodo DOI 10.5281/zenodo.19056919 and 10.5281/zenodo.19375792.
Eureka GCC Semifinalist	2026	Selected as Semifinalist for Eureka GCC Dubai 2026 — competitive pan-GCC innovation program.

Why Now: The Regulatory Window Is Closing

For the first time, AI governance is not a recommendation — it is a legal mandate. Organizations that acquire governance infrastructure now will be compliant by default. Those that wait will face retrofit costs that dwarf the investment.

Framework / Event	Deadline / Status	Implication for Automated Systems
EU AI Act — High-Risk AI	August 2026	Mandatory audit trails, human oversight mechanisms, and explainability for all high-risk AI systems.
NIST AI RMF	Adopted — 2023	De facto standard for US federal agencies and regulated industries. Measurability and governance controls required.
UAE AI Governance Framework	Active — 2025	UAE mandates AI accountability standards across financial services, healthcare, and public sector.
SEC / FCA / MAS Guidance	Published — 2024–2025	Financial regulators across US, UK, and Singapore have published AI governance expectations for trading and credit systems.
ISO/IEC 42001	Published — 2023	International AI management system standard. Requires documented risk controls and incident traceability.

The Founder

Harold Nunes

Solo Founder & CEO — OMNIX

Independent systems architect. While building automated trading systems, Harold identified a gap that no existing product addressed: the absence of pre-execution governance with cryptographic accountability across any domain.

Rather than wait for someone else to build it, he built it himself — from architecture to production deployment, without a corporate engineering team.

What he built: An 11-checkpoint governance pipeline, post-quantum cryptographic signatures (NIST-standardized), multi-domain deployment across trading, Islamic credit, insurance, and robotics — and 4 peer-indexed academic publications documenting the architecture.

"I couldn't find infrastructure that governed automated decisions with cryptographic accountability. So I built it."

Currently seeking: **\$500K pre-seed · Strategic co-founder · First enterprise clients**

Contact & Next Steps

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Three ways to move forward:

1. **Start with Shadow Mode** — free, no disruption, immediate governance visibility.
2. **Schedule a technical demo** — see the pipeline evaluate a real decision live.
3. **Request the full technical specification** — available under NDA for institutional due diligence.

OMNIX is raising \$500,000 USD pre-seed at a \$3,000,000 pre-money valuation. For investor inquiries contact Harold Nunes directly.

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